

**ADMINISTRATIVE INFORMATION****FMI No: 25377****FMI Title: CT MX165 Qj & NP Replacement Tubes**

FMI TYPE	CHARGE CODES	IMPLEMENTATION DATES
☒ Mandatory Safety ☐ Mandatory ☐ Mandatory on Request	GEMS-AM: GEMS-E: GEMS-A:	Issue Date: Close Date:

☒ Staged FMI	Customer List provided by FMI Administration. If you identify additional units within effectivity range, but they are not shown on the customer list, contact FMI Administration in your Pole. ☐ Parts and any special tools are furnished automatically. To address shortages and ordering exceptions, contact FMI Administration in your Pole. ☐ Parts Ordering: GEMS - Americas, contact your Region logistics coordinator GEMS - Europe, Order FMI kits from GPO/EPL Buc GEMS - Asia, Contact your region logistics coordinator ☒ No Parts are required to complete this FMI.
☐ Batch with	FMI # To minimize costs, combine implementation of this FMI with the FMIs listed for all customers where the effectivity is the same (Please see your customer lists).
☐ Non-Staged	No Customer List is provided. Each service location must identify affected units and place part orders as indicated in the Special Instructions.

Technical Support

GEMS - Americas	GEMS - Europe	GEMS - Asia
GE Medical Systems Insite Center (414) 524-5300 800-321-7973 Milwaukee, WI 53201, USA	GEMS - E / ESC BP34 F 78533 BUC CEDEX, FRANCE Tel: (33 1) 39 20 00 07 or ESC Fax: (33 1) 30 70 99 70	Asia Support Center Phone: 81-426-56-0033 Fax: 81-426-48-2905

Special Instructions:**Unit Tracked:****FMI Instructions Part No.: 5176743-100****Do NOT leave unsecured on site.**



Completion Certification Report

Mail To

GEMS - Americas

G.E. Medical Systems
Attn: FMI Admin. W-523
P.O. Box 414
Milwaukee, WI 53201

GEMS - Europe

GEMS - Europe
FMI Admin. 322
BP34
F 78533 Buc Cedex, France

Send to your
Country FMI or Service
Administrator

FMI No.: 25377 FMI Title: CT MX165 Qj & NP Replacement Tubes

Customer Full Name: _____

Number and Street Address: _____

City and Country: _____

Country Code

Site Number

System I.D.

Unit Tracked:

Model Number	Serial Number	Compl. Date DD/MM/YY	FMI Comp. Code	Service Hours	
				Labor	Travel

FMI Completion Code:

- 1.) Modification Complete or previously modified.
- 2.) FMI not required per FMI instruction.
- 3.) Not modified. Unit in storage. FMI kit returned to stock.
- 4.) Modification refused or equipment altered. Customer signature and title required.
- 5.) Unit location unknown or unit not in area. Product Locator notified.

**Customer Refusal or Equipment Altered
- Modification Pending.**

Installer
Comments

Effectivity

Systems with the following serial numbers:

Country	Type	Casing Serial No.	Insert Serial No.	SID	Customer Name
AUSTRIA	D3142T	69371CX3	32126BI5	AP5226CT01	Fink
FRANCE	D3142T	68801CX0	31831BI1	Goldseal	Unknown
	D3142T	68794CX7	28089BI1	M4161310	Arpajon
	D3142T	68628CX7	28443BI0	M4207604	Intercommunal SUD Leman-Valserin
	D3142T	68791CX3	31506BI9	M4170309	Camille Guerin
GERMANY	D3142T	68630CX3	28446BI3	PMAB01CT03	Goecke & Partner
	D3142T	68793CX9	31507BI7	PC0153CT03	Wujciak D. -DiestelHorst M.
GREECE	D3142T	68783CX0	31242BI1	H2077CT02	Unknown
	D3142T	68784CX8	29441BI3	H2126CT03	Unknown
HUNGARY	D3142T	68789CX7	31426BI0	HU1045CT01	HU1045 -CSONGRAD MEGYEI ONKORM. TERULETI
	D3142T	69373CX9	32127BI3	HU1025CT02	Toldy Ferenc Korhaz
ITALY	D3142T	68795CX4	31505BI1	A5355205	S. ANTONIO ABATE OPS. TRAPANI
	D3112T	68828CX3	31579BI6	A148242801	CME,Radiologia Medica Massa Srl Caserta
	D3112T	68829CX1	31580BI4	A5302306	S. Martino O.C. Oristano
	D3142T	68797CX0	31737BI0	A5251306	A52513 -BIOCONTROL IMAGING SRL
	D3142T	69365CX5	32586BI0	A5636001	Centro Vomero-Dr. Basile
	D3142T	69368CX9	32587BI8	A5250905	S. Michele O.C.
LATVIA	D3142T	69086CX7	31832BI9	LATVIA	LATVIA
Morocco	D3142T	68790CX5	31433BI6	MA1294CT01	Hopital Alid Rissa
Netherlands	D3142T	68800CX2	31582BI0	TLB0120	Dr. Bernard Verbeeten Ins
POLAND	D3112T	68683CX2	27825BI9	600036CT02	Wojewodzki Szpital Zespolony
Portugal	D3142T	69083CX4	31827BI9	PPR14202	PPR142 -SANTAC
	D3112T	69552CX8	32125BI7	PPR16003	PPR160 -CL DE GONDOMAR
	D3112T	68826CX7	31577BI0	PPR47100	SCM Benavente
SPAIN	D3142T	68799CX6	31671BI1	EE214305	Obispo Polanco
	D3112T	68841CX6	31825BI3	EP591305	EP5913 -REINA SOFIA
	D3142T	68792CX1	31512BI7	MP870803	MP8708 -CARMEN RODRIGUEZ GUIRADO
SWISS	D3112T	68838CX2	31820BI4	502010CT02	502010 -INSELSPIITAL BERN
	D3142T	68785CX5	31246BI2	730060CT01	Spitalzentrum Oberwallis

Purpose

The purpose of this FMI is to replace the x-ray tube for bolt issue.

Prerequisite FMIs

None.

Related FMIs

None.

Furnished Materials

None.

Required Tools

Description
Hoist 46-195700G3
Slicone Oil
Candle Stick HV Lead Wrench
Tie-wraps
Kim Wip
Functional Check / Adjustment Manual (CD-ROM)

People Power

One Field Engineer for 8 hours plus travel.

Section 1.0 Procedure

Please verify that the serial numbers of the tube mounted on the system, correspond with the serial numbers mentioned in the FMI. If the tube is NOT the tube which is listed in the FMI document, tube replacement is not necessary. Please contact OLC or your CT-modality leader in this case.

Else, perform tube-change procedure as per latest GEYMS CT Service Documentation CD-ROM (p/n 2254979). For your reference, a copy of this procedure is included in this document.



NOTICE

For systems powered from a power distribution box which has a lockout capability, perform the following steps at appropriate timings in each component replacement procedure:

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Turn off and lock out the system power at the main input (Power Distribution Box).

.....

Restore system power.

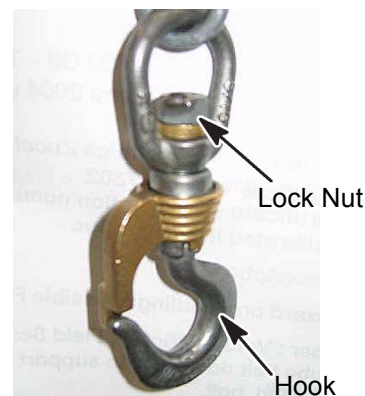
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WARNING



IF THE HOIST IS USED WITH THE LOOSE HOOK OF THE LOCK NUT, THE X-RAY TUBE MAY DROP FROM THE HOIST AND A SERVICE PERSON MIGHT BE SERIOUSLY INJURED AS A RESULT. CHECK IF THE LOCK NUT IS TIGHT ENOUGH PRIOR TO THE USE. IF IT IS LOOSE, DO NOT USE THE HOIST, THEN ORDER THE NEW HOIST.



CAUTION



BURN/SHOCK HAZARD.

Wait more than fifteen minutes following the last scan before performing the following procedure.

1.1 For ProSpeed System (X-Ray Tube: D3112T)

Note: For the Gantry 2175xxx or later (Front Access Gantry) with Compact Siting Mode 2 (limited in table down movement), X-ray tube replacement requires two service engineers.

- 1.) Understand and follow the Safety Guidelines Manual (P9011AW).
- 2.) Switch OFF the "Rotate", "XG" and "Slip Ring 115V" switches.
- 3.) Remove all Gantry covers (Front, rear and both sides).
- 4.) Switch OFF the front XG switch located beside the Detector.
- 5.) Rotate the Gantry until the Tube is approximately 90
- 6.) Insert the Azimuth lock pin.
- 7.) For the Gantry: 2175xxx or later (Front Access Gantry)

- a.) Remove the Hoist Rail locking screw.

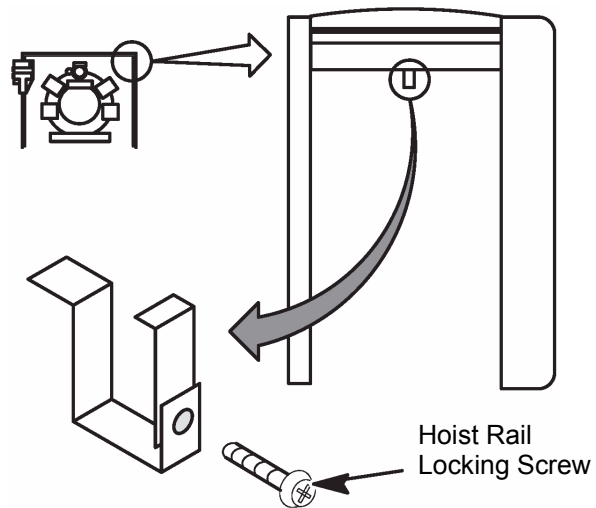


Figure 1 - Hoist Rail Locking Screw

- b.) Slide out the Hoist Rail, then install the hoist as shown.

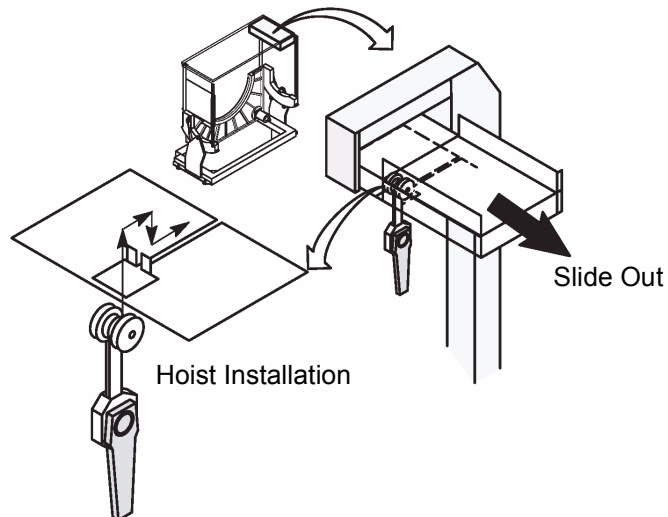


Figure 2 - Hoist Installation

- c.) Slide the Hoist Rail to appropriate position, then attach the hoist chain to the tube hook and pull in all loose chain until the chain is taut.

- 8.) For the Gantry 2137xxx or lower / P9183xx (Conventional lateral Access Gantry).
 - a.) Remove the front corner support pillar by unscrewing its three Allen screws. (Figure 3)
 - b.) Install the hoist (and stopper for MX165Zj) as shown in Figure 3.
 - c.) Secure the hoist chain to the tube hook and pull in all loose chain until the chain is taut. (Figure 3)

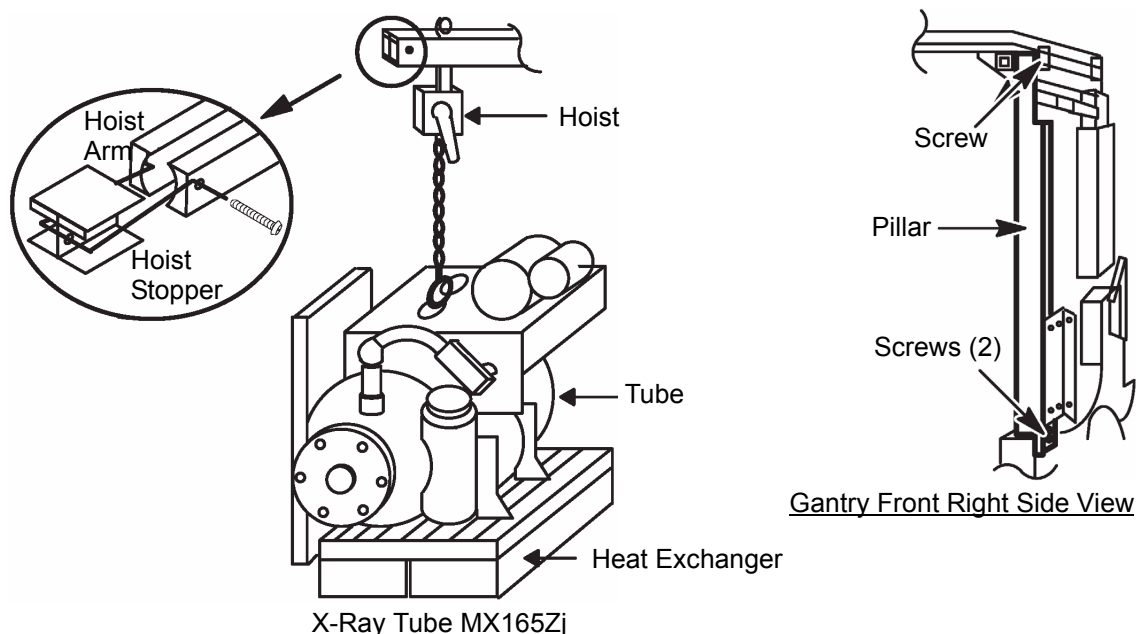


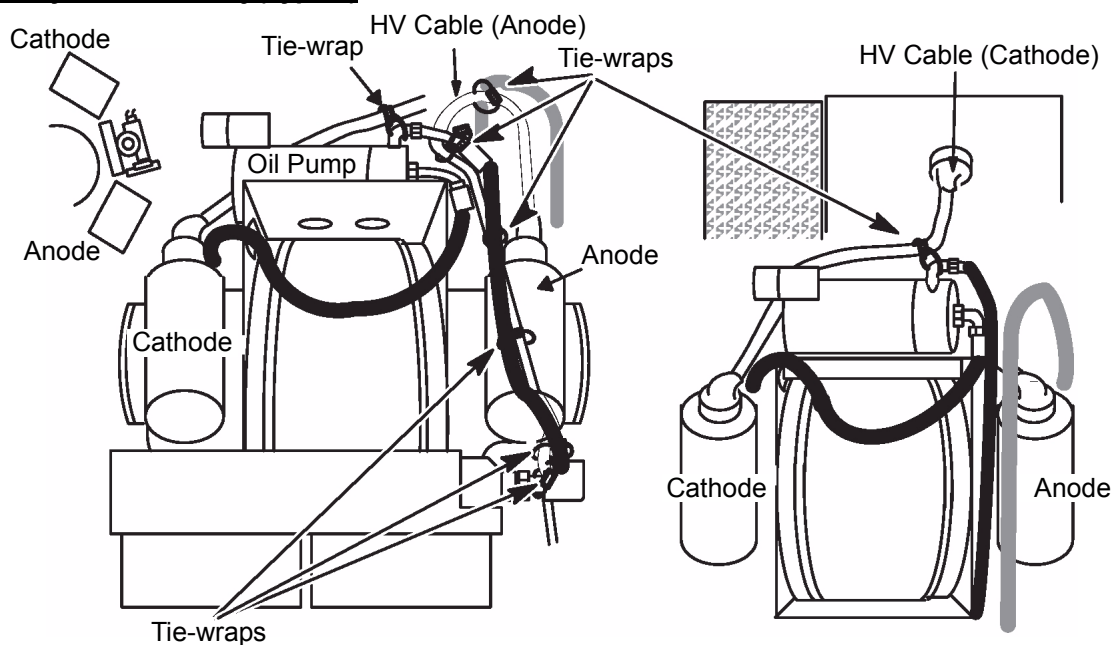
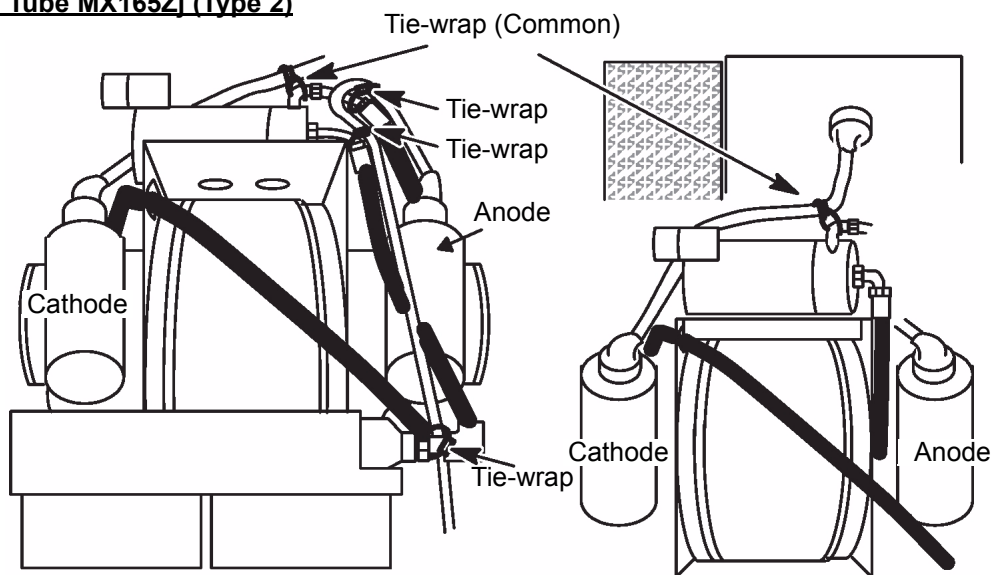
Figure 3 - Hoist Pillar

- 9.) Remove the Filter cover and the Bow.Tie Filter. Refer to Section 1.15, COLLIMATOR FILTER of tab 2, TABLE/GANTRY COMPONENT REPLACEMENT.
- 10.) Pull out the Cathode main HV cable lead from the Tube and discharge it by touching all three prongs to a ground such as the tank.
- 11.) Wipe any residual oil from the lead to avoid dripping and insert kimwipe into the receptacle to prevent oil from leaking.

Note: Take care not to damage the HV CABLE leads.

- 12.) Pull out the Anode main HV cable lead from the Tube and discharge it by touching all three prongs to a ground such as the tank.
- 13.) Wipe any residual oil from the lead to avoid dripping and insert kimwipe into the receptacle to prevent oil from leaking.
- 14.) Cut the four tie wraps shown in Figure 4, 3 Anode and 1 common.
Two types of MX165Zj X-ray tube exists in the field (same P/N). One is the oil tube from cathode side connected directly to the oil pump (Type 1), another is the tube connected to the radiator (Type 2).

Note: For the Gantry 2175xxx (Front Access Gantry), only the Type.2 tube can be installed.

X-Ray Tube MX165Zj (Type 1)**X-Ray Tube MX165Zj (Type 2)****Figure 4 - Tube/Tank Configuration for X-Ray Tube MX165Zj**

- 15.) Disconnect the rotor control connector.
- 16.) Disconnect the wire connector on the cathode side ([Figure 5](#)).
- 17.) Unscrew the M6 cap screw, which is used for ISO adjustment ([Figure 5](#)).

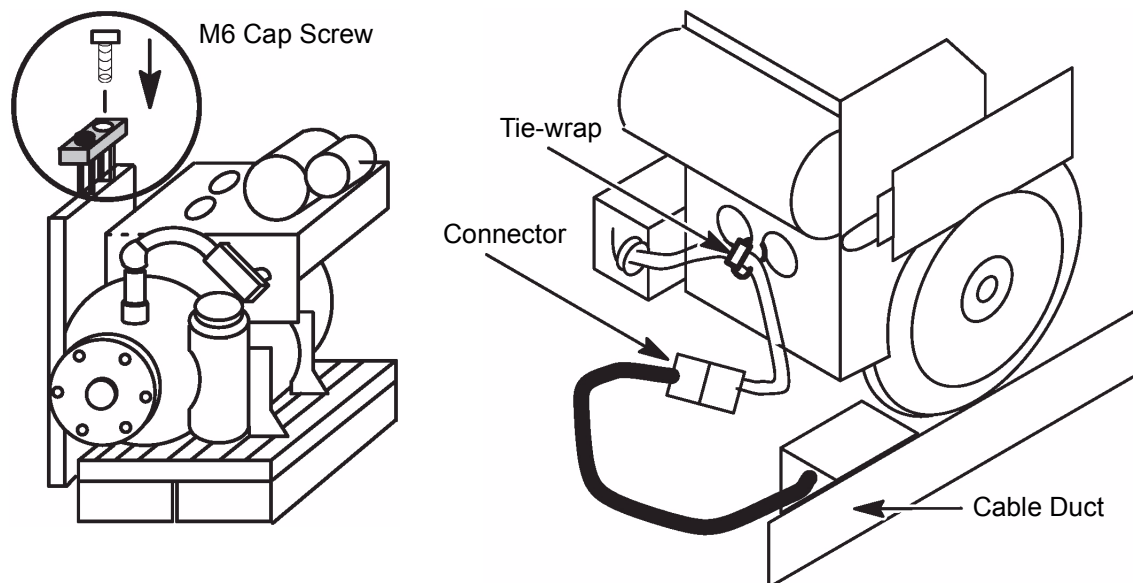


Figure 5 - X-Ray Tube Front View and M6 Cap Screw

- 18.) Loosen and remove the four mounting screws.

Carefully with the tube secure: Torque = 20 lbf.ft | 28 Nm | 285 kg.cm

- 19.) Remove the mounting plate.
- 20.) Dispose of the four plate screws, do not use them for the new tube.
- 21.) For the Gantry: 2175xxx or later (Front Access Gantry)
Use the hoist to slowly lower the tube to the floor, with the mounting plate surface facing down.
- 22.) For the Gantry 2137xxx or lower / P9183xx (Conventional lateral Access Gantry)
Use the hoist to swing the tube to the side and slowly lower it to the ground with the mounting plate surface facing down.
- 23.) Install the mounting plate onto the new tube. (Torque = 20 lbf.ft / 28 Nm / 285 kg.cm)
- 24.) Use the hoist to raise the new tube into place.
- 25.) Replace the Tube and perform steps 18 to 6 in their reverse order.
In [step. 18.](#), use new screws (U0442AA) for mounting the Tube to the Gantry.
- 26.) Perform a clearance check. (Refer to Functional Check & Adjustment, Gantry, Rotational Mechanism, Clearance between Rotational and Stationary Components.)
- 27.) Perform the Rotation Balance check/adjustment after replacement of the HV Tank. (Refer to Functional Check & Adjustment, Gantry, Rotational Mechanism, Rotation Balance.)
- 28.) Perform the following check/adjustment procedures after the Tube replacement according to Functional Check & Adjustment:
- | | |
|---------------------|---|
| * X-RAY ALIGNMENT | System, X-ray Alignment |
| * DAS OFFSET CHECK | DAS/Detector, DAS Operation, DAS Offset Data |
| * mA CALIBRATION | X-Ray Generator, X-ray Tube, Tube Calibration |
| * KV & MA CHECK | X-Ray Generator, KV and MA |
| * DAS COUNT CHECK | X-Ray Generator, X-ray Tube |
| * CALIBRATION | System, System Calibration |
| * CT NO. ADJUSTMENT | System, System Calibration |
| * IMAGE PERFORMANCE | System, Image Performance Verification |
- 29.) Install all Gantry covers (Front, rear and both sides).

1.2 For HiSpeed System (X-Ray Tube: D3142T)

Understand and follow the Safety Guidelines Manual.

- 1.) Set the Gantry to the 0° tilt and the HOME position.
- 2.) Switch OFF the "Rotate", "XG Power" and "Slip Ring115V" switches at the rear Gantry base.
- 3.) Remove the front, right side, and left maintenance covers.
- 4.) Rotate the Gantry until the x-ray tube is at the 90 degree position.
- 5.) Insert the azimuth lock pin.
- 6.) Remove the Hoist Rail locking screw.
- 7.) Slide out the Hoist Rail, then install the Hoist as shown in [Figure 6](#).
- 8.) Slide the Hoist Rail to appropriate position, then attach the hoist chain to the X-ray tube hook and pull in all loose chain until the chain is taut.

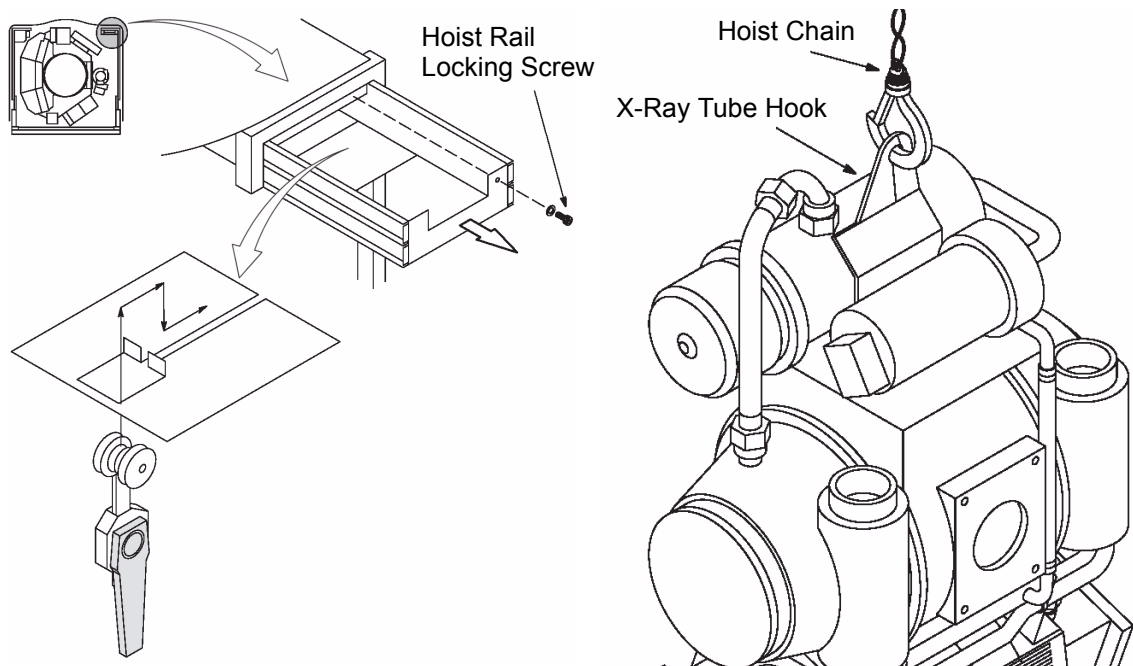


Figure 6 - Hoist Installation

- 9.) Pull out the Cathode main HV cable lead from the X-ray tube and discharge it by touching all three prongs to a ground such as the tank.

Note: Take care not to damage the HV Cable leads.

- 10.) Wipe any residual oil from the lead to avoid dripping and insert kimwipe into the receptacle to prevent oil from leaking.
- 11.) Pull out the Anode main HV cable lead from the X-ray tube and discharge it by touching all three prongs to a ground such as the tank.
- 12.) Wipe any residual oil from the lead to avoid dripping and insert kimwipe into the receptacle to prevent oil from leaking.
- 13.) Cut the tie wraps holding the HV cables to the X-ray tube housing.
- 14.) Disconnect the rotor control connector.
- 15.) Disconnect Fan/Pump power cable connector.

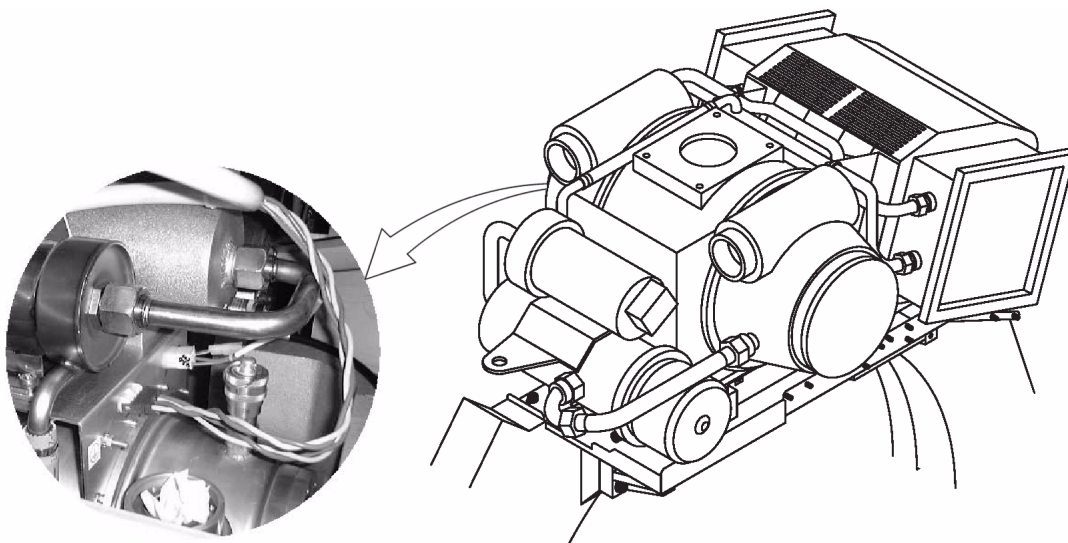


Figure 7 - External Cable Connector

- 16.) Unscrew the long screw, which is used for ISO adjustment.
- 17.) Loosen and remove the four mounting screws.
(Torque all four (4) bolts to the following specification at installation.)

20 ft-lbs

28 Nm

285 kg-cm

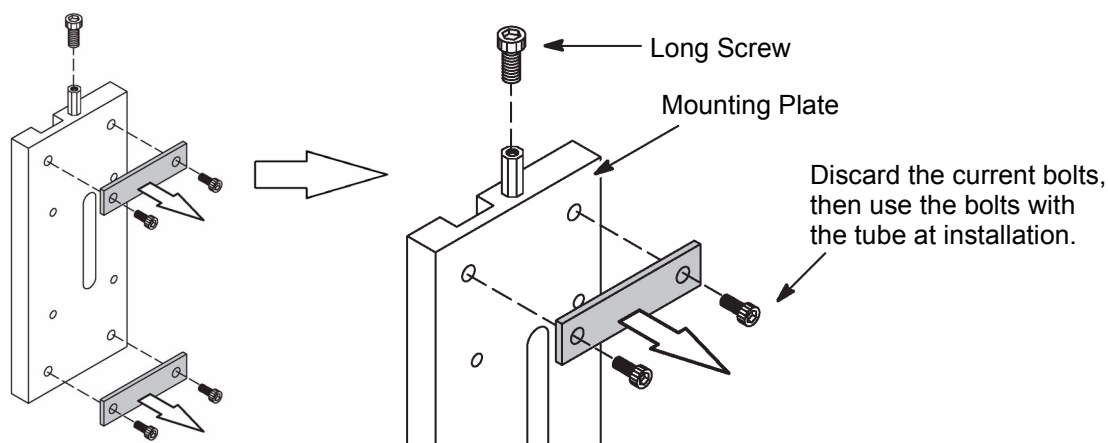


Figure 8 - X-Ray Tube Removal

- 18.) Use the hoist to slowly lower the X-ray tube to the ground.
- 19.) Remove the mounting plate.
(Dispose the four plate screws, do not use them for the new X-ray tube.)
- 20.) Install the mounting plate onto the new X-ray tube. (Torque: 20 ft.lbs / 28 Nm / 285 kg.cm)
- 21.) Use the hoist to raise the new X-ray tube into place. Then Install the new X-ray tube by performing [step. 17.\) through step. 6.\)](#). In the [step. 17.\)](#), use new screws for mounting the X-ray tube to the Gantry.
- 22.) Re-tie any tie wraps which were removed in [step. 13.\)](#). (see illustration below).

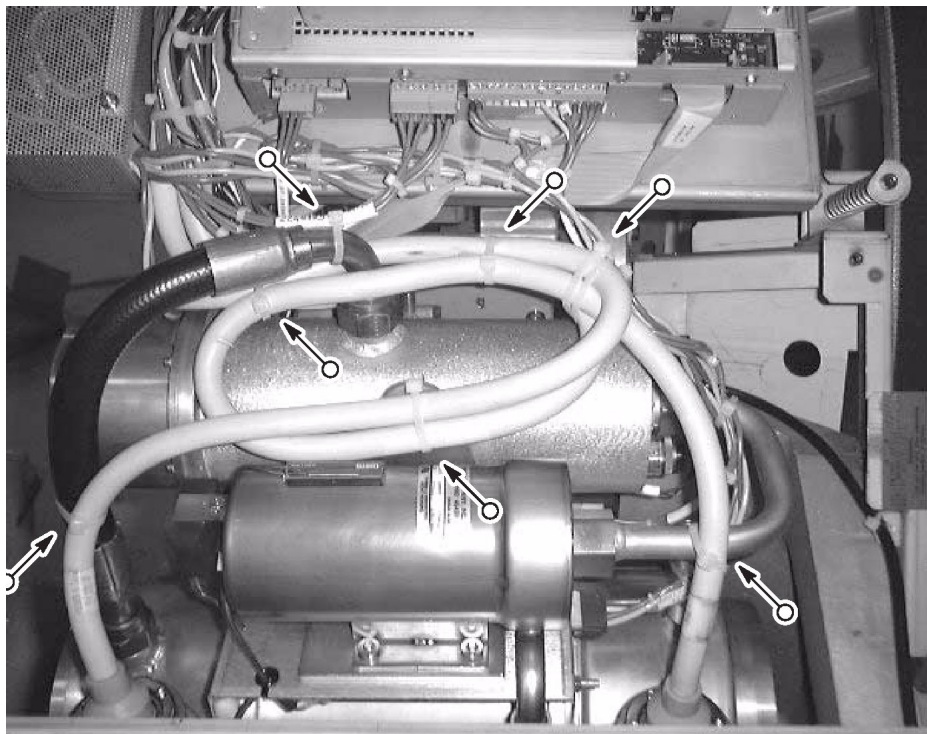


Figure 9 - X-Ray Tube Tie-Wrap Configuration

- 23.) Switch ON the "Slip Ring 115V" switch at the rear Gantry base, and verify that the radiator FAN for the x-ray tube cooling and JEDI Power Unit FAN are operating normally.
- 24.) Update 'Tube Usage' Log and 'Tube Spits' log:
 - a.) Select **Update System Log**; refer to below:
Service Menu -> Update System Log
 - b.) Select **Tube Usage** from the lists,
 - c.) Enter Tube **Insert** Serial Number (Not Housing Serial Number), and press [Create].
 - d.) Select **Tube Spits** from the lists, and press [Create].
- 25.) Perform the following to initialize the Filament Data:
 - a.) Select the '**Clear Filament aging/Tube status**' menu as below;
Service Menu -> Generator Test -> Clear Filament aging / Tube stats
 - b.) Click '**Apply**', then click '**OK**' to clean up the old Filament/Tube data.
The system automatically restarts.

26.) Perform the following check/adjustment according to CD.ROM manual, Functional Check & Adjustment.

- * CLEARANCE BETWEEN ROTATIONAL AND STATIONARY COMPONENTS
- * X-RAY ALIGNMENT
- * KV AND mA CHECK
- * DAS COUNT CHECK
- * Q CALIBRATION (Q.Cal)
- * SYSTEM CALIBRATION
- * CT No. ADJUSTMENT
- * IMAGE PERFORMANCE

Note: If "0402H mA scale error" occurs, run following exposures to allow filament aging correction.

- Stationary, 120 kV 30 mA, small focus, 1 mm x 20 times 1 sec scan 1sec ISD
- Stationary, 140 kV 30 mA, small focus, 1 mm x 20 times 1 sec scan 1sec ISD
- Stationary, 80 kV 60 mA, small focus, 1 mm x 20 times 1 sec scan 1sec ISD
- Stationary, 120 kV 30 mA, large focus, 10 mm x 20 times 1 sec scan 1sec ISD
- Stationary, 140 kV 30 mA, large focus, 10 mm x 20 times 1 sec scan 1sec ISD
- Stationary, 80 kV 60 mA, large focus, 10 mm x 20 times 1 sec scan 1sec ISD

